Step 1: Importing Libraries

```
import pandas as pd
from sklearn.cluster import KMeans from sklearn.preprocessing import StandardScaler import
matplotlib.pyplot as plt
```

Step 2: Creating the Dataset

```
# Example dataset
data = {
    'age': [25, 34, 45, 31, 40, 35, 28, 50], 'annual_income': [15, 18, 20, 22, 24, 25, 30, 35],
'spending_score': [39, 81, 6, 77, 40, 76, 94, 3]

                                                                                    }

df = pd.DataFrame(data)
```

Step 3: Standardizing the Data

Since K-Means relies on Euclidean distance, it is essential to standardize the features to ensure equal contribution: scaler = StandardScaler()

```
scaled_data = scaler.fit_transform(df)
```

Step 4: Applying K-Means Clustering

Cluster the data into three groups: kmeans = KMeans(n_clusters=3, random_state=42) kmeans.fit(scaled_data)

```
df['cluster'] = kmeans.labels_
```

Step 5: Visualizing the Clusters

```
plt.scatter(df['annual_income'], df['spending_score'], c=df['cluster'], cmap='viridis')
plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.title('Customer Segmentation') plt.show()
```

Insights

1. **Cluster Interpretation:**
    - Each cluster represents a distinct group of customers, such as high-income low-spenders, low-income high-spenders, or moderate-income moderate-spenders.
2. **Actionable Strategies:**
    - High-income low-spenders can be targeted with premium offers to encourage spending.